

Fermat's Theorem

Module I

Fermat's theorem

If p is a prime number and a is a positive Integer not divisible by p then $a^{p-1} \equiv 1 \pmod{p}$, where p is a prime number and $p \nmid a$.

That can be re-written as,

$$a^{p-1} \pmod{p} \equiv 1 \pmod{p}, a=2 \text{ and } p=3.$$

$$\rightarrow a^{p-1} \pmod{p} \equiv 1$$

$$\rightarrow 2^{3-1} \pmod{3} \equiv 1 \pmod{3}$$

$$\rightarrow 4 \pmod{3} \equiv 1 \pmod{3}$$

$$\rightarrow 1 \equiv 1$$

Fermat's theorem

Example1:

Lets take $a=2$ and $p=11$.

$\rightarrow a^{p-1} \bmod p \equiv 1$, , where p is a prime number and $p \nmid a$.

$\rightarrow 2^{11-1} \bmod 11$ [11 is a prime number and 2 is not divisible by 11]

$\rightarrow 2^{10} \bmod 11$

$\rightarrow 1024 \bmod 11 = 1$

Example2:

Lets take $a=213$ and $p=523$.

$a^{p-1} \bmod p \equiv 1$, , where p is a prime number and $p \nmid a$.

$\rightarrow 213^{522} \bmod 523$ [523 is a prime number and 213 is not divisible by 523]

$\rightarrow 1$